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MILITARY SPECIFICATION

CONVERTERS, SYNCHRO SIGNAL (PANEL MOUNTED) GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the synchro signal converters (hereinafter referred to as converters) used for: (1) increasing the capacity of a synchro signal transmission system, (2) converting synchro transmission from either 60 to 400 hertz (Hz) or 400 to 60 Hz synchro signal transmission, (3) converting relative bearing signals to true bearing signals, (4) changing synchro speeds, and (5) isolating reflected oscillations from the input synchro bus.

1.2 Classification. Converters are classified according to the following characteristics, as specified (see 6.2 and 6.7).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.1 Number and gear ratios of input and output channels. The number and gear ratios of the input and output channels identified as specified in table I.

1.2.2 1/18 speed output accuracy. The accuracy of the 1/18 speed output of triple output converters identified as follows:

(a) 0 - Not applicable
(b) 1 - 0.2 degree accuracy
(c) 2 - 0.5 degree accuracy

1.2.3 Input and output voltages and frequencies. The input and output voltages and frequencies identified as specified in table II.

1.2.4 Size and type of output synchros. The size and type of output synchros identified as specified in table III.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

FF-B-171	- Bearings, Ball, Annular (General Purpose).
FF-B-185	- Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning.
FF-B-187	- Bearing, Roller, Tapered.
QQ-A-200	- Aluminum Alloy, Bar, Rod, Shapes, Structural Shapes, Tube and Wire, Extruded: General Specification for.
QQ-A-225	- Aluminum and Aluminum Alloy Bar, Rod, Wire, or Special Shapes; Rolled, Drawn, or Cold Finished; General Specification for.
QQ-A-250	- Aluminum and Aluminum Alloy Plate and Sheet: General Specification for.
QQ-C-320	- Chromium Plating (Electrodeposited).

MILITARY

MIL-I-10	- Insulating Compound, Electrical, Ceramic, Class L.
MIL-I-631	- Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-S-901	- Shock Tests, H.I. (High Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-T-7928	- Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification for.
MIL-P-10971	- Pin, Spring, Tubular (Coiled and Slotted).

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MILITARY (Continued)

MIL-F-15160 - Fuses: Instrument, Power, and Telephone.
MIL-F-15160/77 - Fuses (Indicating), Style F77A.
MIL-T-16366 - Terminals, Electrical Lug and Conductor Splices, Crimp-Style.
MIL-W-16878 - Wire, Electrical, Insulated, General Specification for.
MIL-W-16878/4 - Wire, Electrical, Polytetrafluoroethylene (PTFE) Insulated, 200°C, 600 Volts, Extruded Insulation.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts): Packaging of.
MIL-B-17931 - Bearings, Ball, Annular, For Quiet Operation.
MIL-F-19207 - Fuseholders, Extractor Post Type, Blown Fuse Indicating and Non-Indicating, General Specification for.
MIL-F-19207/38 - Fuseholders, Extractor Post Type, Blown Fuse Indicating, Type FHL57G.
MIL-S-19500 - Semiconductor Devices, General Specification for.
MIL-S-20708 - Synchros, General Specification for.
MIL-A-21180 - Aluminum-Alloy Castings, High Strength.
MIL-S-22432 - Servomotors, General Specification for.
MIL-S-23190 - Straps, Clamps, and Mounting Hardware, Plastic and Metal for Cable Harness Tying and Support.
MIL-C-28748 - Connectors, Plug and Receptacle, Rectangular, Rack and Panel, Solder Type and Crimp Type Contacts, General Specification for.
MIL-C-28748/9 - Connectors, Receptacle, Electrical, Rectangular, Rack and Panel, Polarized Center Jackscrew or Guidepin Style, Crimp Type Removable Pin Contacts, Size 16.
MIL-C-39022 - Capacitors, Fixed, Metallized, Paper-Plastic Film, or Plastic Film Dielectric, Direct and Alternating Current, (Hermetically Sealed in Metal or Ceramic Cases), Established Reliability, General Specification for.
MIL-T-55164 - Terminal Boards, Molded, Barrier, Screw and Stud Types, and Associated Accessories, General Specification for.

(See supplement 1 for list of associated specifications.)

STANDARDS

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited).
MIL-STD-242 - Electronic Equipment Parts Selected Standards.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Emissions and Susceptibility.

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MILITARY (Continued)

MIL-STD-462 - Measurement of Electromagnetic Interference Characteristics.
MIL-STD-701 - List of Standard Semiconductor Devices.
MIL-STD-710 - Synchros, 60 and 400 Hertz, Selection and Application of.
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1303 - Painting of Naval Ordnance Equipment.
MIL-STD-1399, SECTION 300 - Interface Standard for Shipboard Systems, SECTION 300 Electric Power, Alternating Current. (Metric)

HANDBOOKS

MILITARY

MIL-HDBK-225 - Synchros Description and Operation.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

DDS 430-2 - Procedure for Determining Synchro System Loading Capacity.

(Application for copies should be addressed to the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 153 - Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware. (DoD adopted)
B 26 - Standard Specification for Aluminum-Alloy Sand Castings. (DoD adopted)
B 108 - Standard Specification for Aluminum-Alloy Permanent Mold Castings. (DoD adopted)
B 633 - Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel.

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents may also be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements.

3.1.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.1.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Description.

3.2.1 Converter description. Converters shall consist of one or more input devices, and one or more output synchro devices as specified (see 6.2). Input devices, referred to herein as servos, shall be a servtorq or a control transformer with an amplifier unit and a servomotor as specified. Servo unit shall be coupled between the input and output devices directly or by gearing (see figures 1 through 7). The above parts shall be assembled in a panel mounted enclosure as specified on figure 8.

3.2.2 Signal characteristics. The various signal combinations that may be transmitted through a converter shall be identified as specified in 3.2.2.1 through 3.2.2.2.3

3.2.2.1 Single channel converters.

3.2.2.1.1 Single channel converters (see figures 1 and 2) shall provide one of the following output signal to input displacement ratios:

(a) 1 : 1
(b) 1 : 2
(c) 2 : 1

3.2.2.2 Multiple channel converters.

3.2.2.2.1 Single input converters (see figure 3 or 4) shall provide one of the following combinations of output signal to input displacement ratios:

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(a) 1:36, 1:1, and 1:1/18
(b) 36:36, 36:1, and 36:1/18

3.2.2.2.2 Double input - triple output converters (see figure 5) shall provide the following output signal to input displacements ratios:

36:36, 36:1 , 1:1/18 and 1:36, 1:1, 1:1/18

3.2.2.2.3 Double input - double output converters (see figure 6) shall provide one of the following output signal to input ratios:

36:36, 36:1 and 1:36, 1:1

3.2.2.2.4 Double input - triple output converters (see figure 7) shall provide the following output signal to input displacements ratios:

36:36, and 1:1, 1:1/18

3.2.3 Converter signal input impedance. If a CT is used as the input device, the load placed on the CT's rotor by the amplifier unit shall be greater than 10,000 ohms for 60 Hz units and greater than 15,000 ohms for the 400 Hz units. This prevents excessive loading of the converter's signal source transmitters.

3.2.4 Operation. The typical cycle of operation of the converter shall be as follows: A synchro signal shall be transmitted to the input of the converter. If the converter's input device is a CT, an error voltage is generated on the CT's rotor. This voltage is amplified and is used to control a servomotor. The servomotor drives the shaft of the CT along with the output synchros of the converter gearing. When the shaft of the CT is aligned with the synchro input to the converter, the servomotor stops. A servtorq may be used in place of the CT, amplifier, and servomotor.

3.3 General features.

3.3.1 Characteristics. Characteristics not normally incorporated in the design of commercial equipment are required on Naval ships. The primary requirements which are the basis for the design of all Naval equipment shall be as follows:

(a) Maximum reliability for operation under shipboard conditions of heat, moisture, vibration, shock and ship motion as described herein.
(b) Material and design shall be lightweight and compact consistent with reliable shipboard operation.
(c) Maximum accessibility for inspection and maintenance.
(d) Maximum simplicity of design consistent with other requirements described herein.
(e) Maximum resistance to corrosive action, in particular that of moist sea air and ultraviolet radiation.
(f) Maximum operating and maintenance economy.
(g) Maximum interchangeability of parts.
(h) Minimum number of necessary maintenance parts.

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(i) Satisfactory identification of equipment and parts for ease in accurate requisitioning.
(j) Maximum ease of repair by Naval personnel and with the facilities normally found on board ships.
(k) Maximum protection for operating and maintenance personnel.
(l) Maximum ease of installation.

3.3.2 Standard parts. Standard stock parts and hardware shall be used to as great an extent as practicable. For the purpose of this specification, standard stock is defined as material listed in the Federal Supply Catalog and includes such items as bearings, grease, oil, wire, bolts, screws, nuts, and washers. Proprietary parts or parts available from one source shall not be used where equivalent or similar parts are available as standard parts or are available from two or more sources.

3.3.3 Interchangeability. Parts used in the equipment shall be interchangeable to the greatest extent practicable, in accordance with requirement 7 of MIL-STD-454.

3.4 Materials and processes. Materials and processes used in the manufacture of the converters shall conform to the specifications described herein, as well as the following requirements of MIL-STD-454.

(a) Requirement 3 - Flammable materials.
(b) Requirement 4 - Fungus-inert materials.
(c) Requirement 11 - Insulating materials, electrical (see 3.5.4 and 3.6.10.2).
(d) Requirement 13 - Structural welding.
(e) Requirement 15 - Ferrous alloys, corrosion-resistant (only austenitic stainless steel shall be used).
(f) Requirement 17 - Printed wiring.
(g) Requirement 21 - Castings.
(h) Requirement 26 - Arc-resistant materials.
(i) Requirement 43 - Lubricants.

3.4.1 Toxic materials. Radioactive materials, mercury, and heavy metal biocides shall not be used. Materials used in the construction of the converter shall not produce corrosive, deleterious, or toxic fumes or vapors under all conditions herein.

3.4.2 Wood. Wood shall not be used.

3.4.3 Aluminum. Where the use of lightweight corrosion-resistant metal is desired, wrought aluminum alloys of the 5000 series shall be used insofar as practicable. These aluminum alloys shall be accordance with QQ-A-200, QQ-A-225, or QQ-A-250. In general, aluminum alloy castings shall be in accordance with alloy 360 or to alloy 356 of ASTM B 108 or ASTM B 26. High strength aluminum alloy castings shall be in accordance with alloy A 356 of MIL-A-21180.

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3.4.4 Magnesium. Magnesium or magnesium alloys shall not be used.

3.4.5 Nonferrous materials except aluminum. Nonferrous materials, except aluminum, shall conform to commercial standards and shall be of a quality to meet the requirements of this specification.

3.4.6 Zinc. Zinc shall not be used where the increased electrical resistance of the surface, due to the protective treatment, has a harmful effect on electrical performance.

3.4.7 Ceramics. Ceramics shall not be used for structural parts. Ceramics, where used for other applications, shall be in accordance with MIL-I-10.

3.4.8 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and shall be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified.

3.4.9 Painting. After machining, welding, and brazing operations have been completed, the front panel of the converter enclosure shall have rust or other visible corrosion products removed and shall be thoroughly cleaned of grease, oil, and dirt, and painted in accordance with system 37 of MIL-STD-1303.

3.4.10 Protection against corrosion. In order to prevent corrosion, metal parts shall be of corrosion-resisting materials or other materials treated as specified in 3.4.10.1 to render them adequately resistant to corrosion. Internal parts fabricated of nonferrous materials contained in a watertight enclosure shall not be given a corrosion-resistant treatment.

3.4.10.1 Corrosion-resisting treatments. The most applicable corrosion-resisting treatment of those listed below shall be applied after fabricating operations such as welding, machining, drilling, and tapping have been completed. Corrosion-resisting treatments shall not be applied to surfaces which are in functional contact where gouging or binding may be a factor, where the treatment might interfere with the normal functioning, or where electrical grounding through the surface is required.

(a) Zinc coating, hot dip galvanizing in accordance with ASTM A 153, for parts other than threaded fasteners, unless part specifications contain other requirements.
(b) Electroplating of zinc in accordance with type II, Fe/Zn 13 of ASTM B 633, for surfaces not to be painted except that class 3 thickness may be used for externally threaded parts, bolts, studs and washers. Type II, class 1 plating may be used on parts which

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are continuously exposed to temperatures in excess of 150 degrees Fahrenheit (°F) or are intermittently exposed for short periods to temperatures of approximately 300°F. Either type II, class 2 or type III, class 2 coating shall be used for surfaces that are to be painted.

(c) Electroplating of chromium in accordance with QQ-C-320.
(d) Chemical film in accordance with class 1A of MIL-C-5541.

3.4.10.2 Selection of metals. In order to minimize corrosion attack due to electrolytic action between dissimilar metals when used in contact with each other, selection shall be in accordance with MIL-STD-889.

3.5 Mechanical parts and features.

3.5.1 Mechanical requirements. Mechanical parts and features of the converters shall conform to the following requirements of MIL-STD-454.

(a) Requirement 12 - Fastener hardware (see 3.5.1.1).
(b) Requirement 21 - Castings.
(c) Requirement 63 - Special tools.

3.5.1.1 Fastener hardware. Thread cutting and self-tapping screws shall not be used. A lock washer shall be installed under the head of other than flat-head machine screws. Flat-head machine screws shall be secured with a staking compound.

3.5.2 Bearings. Bearings shall be in accordance with requirement 6 of MIL-STD-454 and as specified in 3.5.2.1 through 3.5.2.4.

3.5.2.1 Ball bearings. Ball bearings shall be in accordance with FF-B-171. They shall be applied and installed in accordance with the requirements of that specification. Grade 00 ball bearings shall be used wherever possible.

3.5.2.2 Roller bearings. Roller bearings shall be in accordance with FF-B-185 or FF-B-187 and shall be applied and installed in accordance with the requirements of this specification.

3.5.2.3 Sleeve type bearings. Sleeve type bearings shall not be used.

3.5.2.4 Noise-tested bearings. Where required, noise-tested bearings shall be in accordance with MIL-B-17931.

3.5.3 Rounded corners and edges. Corners and edges of external surfaces, normally exposed or to be painted, shall be rounded.

3.5.4 Protection of wiring. Edges of parts that will come into contact with wiring shall be rounded or provided with rounded insulating guards, grommets, or pads to prevent damage to the internal wiring (see 3.6.10.4). Stand-offs or protective plates shall be provided where required to prevent wiring from being damaged by terminal nuts or screw threads. Insulating guards, grommets, pads, sheaths, stand-offs, and protective plates shall be in accordance with type F, grade A of MIL-I-631.

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3.5.5 Internal subassembly protection. Complete unit internal subassemblies shall be provided with means to prevent injury to pointers, dials and other parts when the subassembly is removed from its enclosure and rested on a work bench on either its top or bottom.

3.5.6 Drilled or tapped holes. Drilled or tapped holes shall be countersunk to remove burrs and sharp edges. Tapped holes in aluminum shall be fitted with helical inserts.

3.5.7 Size. Converter shall be as specified herein, and as shown on figure 8.

3.5.8 Mounting. Converter shall be capable of being mounted as shown on figure 8. Converter and front panel shall have the necessary strength and rigidity to prevent contact between the converter and the walls of the cubicle under the conditions of shock and vibration (see 3.8.13 and 3.8.14).

3.5.9 Handles. Two handles shall be provided. Handles shall be secured to, but detachable from, the front panel. Handles shall not extend more than 2-1/2 inches from the front panel (see figure 8). Figure 8 shall be used for guidance in the location of the handles.

3.5.10 Accessibility. Parts and subassemblies of the converter shall be easily accessible for repair and adjustment. Adjustment of synchros shall not require disassembly of the converter. Supporting structures shall be provided on the converter to allow repair work without having the unit resting on wiring, electrical connections, or parts that may be dislocated or damaged. Requirement 36 of MIL-STD-454 shall be used as guidance with respect to accessibility.

3.5.11 Gearing. One gear train shall connect the servomotors, receivers, and transmitters. Gears shall provide smooth, continuous tooth contact without interferences, tight spots, loose spots, or other irregularities. Every precaution shall be taken to reduce tooth pressure and backlash to a minimum. Gears fitted on shafts shall be secured thereto by means which will effectively prevent relative motion between the gear and the shaft (see 3.5.12). Gears shall be metallic.

3.5.12 Shaft locking devices. Positive locking devices shall be used to secure collars, gears, and similar parts to their associated shafts. Acceptable methods shall be in accordance with MIL-P-10971 or requirement 28 of MIL-STD-454. Taper pins shall not be used.

3.5.13 Weight. Weight of the converter shall be kept to a minimum.

3.5.14 Mechanical connections. Connections shall be supported in a manner which will preclude breakage and minimize changes in the performance due to vibration, inclination, or shock encountered during use.

3.6 Electrical parts and features.

3.6.1 Electrical requirements. Electrical parts and features of the converters shall conform to the following requirements of MIL-STD-454:

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(a) Requirement 2 - Capacitors.
(b) Requirement 5 - Soldering (also covers mounting of electrical parts) (see 3.7.1.6).
(c) Requirement 14 - Transformers, inductors, and coils.
(d) Requirement 17 - Printed wiring.
(e) Requirement 18 - Derating of electronic parts and materials.
(f) Requirement 24 - Welds, resistance, electrical connections.
(g) Requirement 30 - Semiconductor devices.
(h) Requirement 33 - Resistors.
(i) Requirement 69 - Internal wiring practices (see 3.6.8).

3.6.2 Safety, ground potential, and grounding. Safety and grounds shall be in accordance with requirement 1 of MIL-STD-454.

3.6.3 Electronic assemblies. Electronic assemblies shall be compact solid state units and shall not be potted. (Electronic assemblies internal to a device shall be potted, as required.) Plug-in type assembly devices are preferred and shall be used where practicable.

3.6.4 Fuses and fuseholders.

3.6.4.1 Fuses. Fuses shall be style F77A in accordance with MIL-F-15160 and MIL-F-15160/77 and shall be provided and installed as specified (see 6.2). Fuses under 1 ampere rating shall not be used. Fuse size shall be indicated by marking adjacent to each fuse.

3.6.4.2 Fuseholders. Fuseholders shall be type FHL57G in accordance with MIL-F-19207 and MIL-F-19207/38. The fuseholders shall be mounted on the front panel (see figure 8) or as specified (see 6.2).

3.6.4.3 Blown fuse indicator circuit. Blown fuse indicator circuit used with type FHL57G fuseholders shall be energized as specified (see 6.2).

3.6.5 External electrical connections. External electrical connections to the converter shall be made by either a terminal board (type 10TB12 or 6TB24 in accordance with MIL-T-55164) or a connector (part number M28748/09F0N01A in accordance with MIL-C-28748 and MIL-C-28748/9) mounted on the rear of the converter (see 6.2).

3.6.6 Synchro signal selector. A synchro signal selector shall be provided in multispeed converters (for example, 1 and 36 speed; 1, 18, and 648 speed) in order to select the controlling signal from the input synchros. The synchro signal selector is called a speed switching network as shown on figures 5 and 6.

3.6.7 Power supply. Converters shall operate within the specified accuracy limits (see 3.7.1) from a type I or II shipboard 115 volts alternating current (Vac) power source, in accordance with MIL-STD-1399, section 300. Power required within the amplifiers shall be obtained from internal sources energized from the synchro reference power supply. Rectification from ac to dc shall be employed in the power supply design to prevent damage to the supply itself, as well as electronic assemblies powered by the supply (see 3.7.1.3 and 3.8.8).

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3.6.8 Insulation resistance clearance. Clearances between any two electrical circuits or between any electrical circuit and ground (metal enclosure or chassis) shall be such as to meet the test conditions for insulation resistance specified in 4.6.16. Insulation resistance of the circuits test shall be not less than 10 megohms when tested as specified in 4.6.16. Creepage and clearance distances between conductors and other conductors, ground or frame in the equipment shall be in accordance with requirement 69 of MIL-STD-454, and in no case shall be less than necessary to prevent arcing or leakage during operation or service.

3.6.9 Transformers and inductors. Transformers and inductors shall conform to requirement 14 of MIL-STD-454, grade 5, class S, life expectancy X, and shall be selected in accordance with MIL-STD-242. Power and audio input transformers shall have electrostatic shields, and the shields shall be grounded.

3.6.10 Wire. Internal hook-up wire shall be in accordance with MIL-W-16878 and MIL-W-16878/4.

3.6.10.1 Method. The use of preformed cables and wiring harnesses is preferred over the point-to-point method of wiring.

3.6.10.2 Arrangement. Wiring shall be arranged to permit bundling by one or more of the following methods:

(a) Binding - Tape for binding shall conform to requirement 11 of MIL-STD-454.
(b) Sleeving insulation - Sleeving insulation shall conform to requirement 11 of MIL-STD-454.
(c) Tying - Plastic devices for tying shall be in accordance with MIL-S-23190. When the wire bundle is formed and secured, the device shall not loosen under vibration.

3.6.10.3 Color coding. Color coding and marking of wires shall conform to requirement 20 of MIL-STD-454.

3.6.10.4 Protection. Wiring, both fixed and flexible, shall be protected to prevent chafing and abrasion due to vibration or flexing as specified in 3.5.4. Plastic sheaths shall be in accordance with type C of MIL-I-631 and may be similar to a covering known as "zipper tubing". A sealing compound shall be used with this tubing. The use of clamps or wrapping and tying devices (see 3.6.10.2) with integral mounting facilities is preferred for securing fixed wiring. Metal clamps, if used, shall be insulated and shall not form a complete loop around the wiring.

3.6.11 Wire terminal lugs. Except for contacts used with connectors, pressure grip terminal lugs in accordance with figure titled "Type WTG terminals, round or rounded tongue, one bolt hole" of MIL-T-16366, or MIL-T-7928 shall be used. Insulated barrel terminal lugs shall be used. Terminal lugs with projections or "lips" shall be used in all cases where necessary to prevent turning of terminal lugs and shorting due to close clearances. Solder type terminal lugs may be used.

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3.6.12 Printed wiring boards. Printed wiring or printed circuit boards shall conform to requirement 17 of MIL-STD-454. Boards shall permit replacement of parts without special tools or techniques (see 3.7.1.6). Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the offices of the Defense Contract Management Area Operations (DCMAO)). If coatings are used, such coatings shall be easily removable by means of a soldering gun or iron without damage to the printed wiring board. Coatings which emit toxic or harmful vapors or gases under heated conditions shall not be used (see 3.4.1).

3.6.13 Relays. Relays shall conform to requirement 57 of MIL-STD-242, except as otherwise specified in MIL-STD-242. Those selected shall operate directly from the normal power supply of the equipment, within the tolerance characteristics thereof. In addition, relays shall preferably be of the sealed, plug-in type, firmly secured in the mounting sockets. Relays containing mercury in any form shall not be used. The device shall operate as specified under all conditions of usage, shall make firm, low resistance contact, and shall retain their setting in any position of the equipment and under any service condition. Contacts shall be noncorrosive and, where practicable, of the self-cleaning type.

3.6.14 Synchros. Synchros shall conform to MIL-S-20708 and shall be as specified herein. Output synchros shall be selected in accordance with table IV herein or in accordance with MIL-STD-710, and shall be as required for the intended application. MIL-HDBK-225 shall be used for guidance in synchro applications and installation. Load values shall be within the maximum limits as specified in table IV for various converter configuration limits.

3.6.15 Servomotors. Servomotors shall be of the greased, sealed ball bearing type. Selection of 60 and 400 Hz units shall be in accordance with MIL-S-22432.

3.6.16 Prevention of servtorq motoring. Motoring is a phenomenon of the servtorq and synchro receiver continuing to run at a speed which may be up to 100 revolutions per minute (rpm) when the rotor is energized in the absence of a signal input to the stator.

3.6.16.1 Preventive measures. Unless otherwise specified (see 6.2), motoring shall be prevented. Means shall include, for example, a sensing device (such as a relay or solid state device) for removal of a rotor voltage (both R1 and R2 connections) in the absence of a stator input (S1, S2 and S3 terminals). Relays, if selected, shall be as specified in 3.6.13 and in accordance with requirement 57 of MIL-STD-454.

3.6.17 Synchro capacitors. When specified (see 6.2), the correct synchro capacitors shall be provided as part of the converter, delta connected across the stator leads (S1, S2, and S3) of differential synchros and control transformers. Each capacitor element shall be rated for 600 volts direct current (Vdc) for 60 Hz synchros and 1000 Vdc for 400 Hz synchros. The capacitance value of each element shall be determined in accordance with NAVSEA DDS 430-2. Table V illustrates representative values of capacitors for specific synchro units.

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3.6.17.1 Tolerances. Where specified in table V, the synchro capacitors shall have the values noted within plus or minus 10 percent, as variations may occur between units of different origin. Each synchro capacitor shall have the capacitance of each of its three capacitor elements balanced within 1 percent, as measured by the divergence of the capacitance of any one leg from the arithmetical mean of the three. Capacitors shall be in accordance with MIL-C-39022, using MIL-HDBK-225 as guidance, and shall be mounted within the converter enclosure.

3.6.18 Synchro connections and markings. Synchro transmitters shall be connected to the converter terminals and the ship wiring (where either may be marked with typical ship circuit designations, terminal "111" to ship wiring "111", and so forth) in such a manner that an increased value (angular displacement) in the peripheral equipment synchro shaft (counterclockwise rotation) would cause a counterclockwise rotation in a standard synchro receiver connected to the ship wiring through the converter terminals (terminal R1 to ship wiring "110", R2 to "110(R), S1 to "111", S2 to "112", and S3 to "113"). Direction of rotation shall be determined while facing the shaft end of the synchro. Synchro torque receivers and control transformers shall be connected to the converter terminals in such a manner that, with direct connection between the converter terminals and ship wiring, counterclockwise rotation of a standard transmitter having direct connection to the ship wire (R1 to "110, R2 to "110(R)", S1 to "111", S2 to "112", S3 to "113") shall cause counterclockwise rotation of the converter dial or pointer. Conversely, for a synchro that requires clockwise rotation for increasing signal, synchro terminals S1, S2, and S3 shall be connected to the converter terminals corresponding to ship wire numbers 113, 112, and 111, respectively. (See figure 10 and MIL-HDBK-225 for additional connection data.)

3.7 Electronic devices and components.

3.7.1 General requirements.

3.7.1.1 Breakdown voltage ratings. For transistors, the type selected shall have a collector-to-emitter and collector-to-base breakdown voltage rating at least three times the voltage of the circuit of use. For diodes and similar devices (for example, thyristors), except zener diodes, the device chosen shall have a peak inverse or peak reverse voltage rating not less than four times the voltage of the circuit wherein used. Microcircuits (also referred to as integrated circuits and integrated devices) selected for application shall have a supply voltage rating of not less than 1.5 times the voltage in the circuit of use. Voltage ratings of devices shall be increased upward as required, in addition to the use of circuit protection, in the event transient or spike voltages are present in a network (see 3.7.1.3).

3.7.1.2 Current ratings. Current ratings of the semiconductor devices (discrete and integrated) selected for application shall be not less than twice those encountered in the circuit wherein used. Should surge currents be present in a network, the applicable ratings of all devices shall be increased upward as required, with particular attention paid to various surge current ratings expressed by device manufacturers. Surge current ratings of devices shall be not less than twice those encountered in the circuits wherein used. Current-limiting means shall be employed whenever necessary to preclude damage to solid state components, especially if the transients occur in the network (see 3.7.1.3).

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3.7.1.3 Transient protection. Should transient or spike voltages and surge currents be present in a network, the applicable rating of solid state devices shall be increased upward as required. Transistors and microcircuits shall also be protected against transients by suitable means (for example, back-biased diodes to avoid exceeding device breakdown voltages). With respect to transistors and integrated devices, when converters are tested as specified in 4.6.8, no damage to the equipment shall occur and dynamic accuracy shall be as specified in 3.7.1. Circuit analysis shall also consider avoidance of exceeding device power dissipation under transient conditions (see 3.7.1.4).

3.7.1.4 Power dissipation. In audio applications, the instantaneous power required to obtain the desired output power shall in no case exceed one-half of the rated maximum power dissipation of a discrete device. For switching applications, the instantaneous power may not exceed one-half of the rated maximum power dissipation of a discrete device only after, and to the extent that, an analysis will show that the peak instantaneous power and the repetition rate of this peak results in an average power one-half or less of the maximum power dissipation of the discrete device. For microcircuits, the instantaneous power specified above for audio and switching applications shall not exceed one-half the maximum rated power dissipation of the complete microcircuit. For voltage dropping devices, the power dissipation shall be at least 50 percent greater than the maximum design value.

3.7.1.5 Temperature considerations. The effect of the ambient temperature on solid state device ratings and equipment performance shall be determined and considered in the design. The maximum rated function temperature of electronic devices shall not be exceeded under any condition of use. The requirements specified in 3.7.1.1 through 3.7.1.4, 3.7.1.6, and 3.7.1.7 shall be considered in the equipment design. The equipment is intended for use in an ambient temperature range of 40 to 149°F.

3.7.1.6 Mounting. Power devices shall be mounted to allow maximum heat transfer. The desired mounting is directly on its heat sink or on the metal of the chassis. A thermal conductive compound shall be applied to the mating surfaces. Equipment design shall be investigated to determine if other components require such mounting. Parts provided with pigtail leads (for example, various types of diodes, transistors, and integrated circuits, and small resistors and capacitors), other than those mounted on printed wiring boards, shall be secured between solder-type terminals on part mounting boards in accordance with the following:

(a) The arrangement of these parts shall be such as to limit the number of leads terminating at any one terminal to three, except for double turret terminals which may terminate three wire per section (turret).
(b) The attachment and number of leads shall be in accordance with requirement 5 of MIL-STD-454.
(c) The clearance between the soldered connection and the body of the part shall be not less than 3/16 inch.
(d) The basic requirements shall predicate that parts with pigtail leads shall not be mounted by means of the wire leads unless other mechanical support is provided, except parts weighing less than 1/2 ounce may be secured by only their leads.

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(e) The leads shall be short as practicable, keeping in mind the clearance limitation of 3/16 inch noted above, and in no case shall the leads be longer than 1 inch.
(f) Parts having a diameter larger than 1/2 inch shall be securely fastened by one or more clamps.
(g) Kinks or other slacks shall be provided in the mounting leads to permit thermal expansion of the terminal or connection board without stressing the parts.
(h) Parts shall be arranged to permit the replacement of any component without removal of or damage to adjacent parts (see 3.6.1).

3.7.1.7 Thermal considerations. Current shipboard considerations shall require an integrated or self-contained thermal design for electronic assemblies to properly dispose of heat generated therein. Portions of the heat removal path shall be designed with as low a resistance to the flow of heat as possible. Sufficient heat sink shall be provided such that the performance of the equipment shall be as specified at the maximum specified ambient temperature (see 3.7.1.5 and 3.7.1.6). Convection cooling shall be a prime consideration for equipment design. Circuit boards shall be mounted vertically to allow maximum transfer to the circulating air.

3.7.2 Synchro electrical zero and equipment mechanical zero. In order to ensure that synchro components will work properly together in a system, it is essential that they be correctly connected and aligned with respect to each other, and to the device or parent equipment with which they are used. Electrical zero is the reference point for the device or parent equipment using synchro components depending upon the particular application of the equipment involved. Unless otherwise specified (see 6.2), whenever a synchro component is used, either as a transmitter or receiver, the synchro electrical zero and the equipment mechanical zero shall be physically positioned to the same point. Synchros shall be electrically zeroed by a method specified in MIL-HDBK-225 prior to delivery of equipment.

3.7.3 Designation and marking. Identification plates, information plates, and other designation plates and markings of units, subassemblies, and parts shall be in accordance with requirement 67 of MIL-STD-454, except that the reversed etched plates shall not be used. Serial numbers on plates for major parts or subassemblies (for example, synchros and servtorq units) shall be required and shall be assigned sequentially. Plates shall include the manufacturer's code, electrical parameters, item name, type designation, National stock number, and so forth.

3.7.4 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454.

3.7.5 Discrete semiconductor devices. Discrete semiconductor devices, such as transistors and diodes, shall be in accordance with MIL-STD-701 with respect to choice and application and shall be in accordance with MIL-S-19500. Only silicon devices shall be used. Selection shall include the requirements specified in 3.7.1 in the design analysis.

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3.7.6 Microcircuits. Microcircuits or integrated circuit devices shall be selected in accordance with requirement 64 of MIL-STD-454. Selection shall include the requirements specified in 3.7.1 in the design analysis.

3.8 Performance.

3.8.1 Dielectric withstanding voltage. When tested as specified in 4.6.15, the converter shall withstand the test potentials between application points specified in 4.6.15. The converter shall display no evidence of insulation breakdown or of internal arcing nor shall winding leakage current exceed 1 milliampere peak. The leakage current limit shall not include the current drawn by the test equipment capacitance. Immediately after meeting this requirement the converter shall meet the insulation resistance requirements of 3.8.2.

3.8.2 Insulation resistance. When tested as specified in 4.6.2, insulation resistance of electrical circuits shall be not less than 10 megohms.

3.8.3 Synchro connections. Connections and markings specified in 3.6.5 shall be determined by the test specified in 4.6.3.

3.8.4 Input operation. The design specified in 3.6.17 shall be determined as specified in 4.6.4.

3.8.5 Static accuracy. The static accuracy of the synchro converters shall be such that when tested as specified in 4.6.5, the errors shall not exceed those specified in the notes of the applicable figure (see figures 1 through 7 and MIL-C-24105 specification sheets). For multiple inputs, "input speed" shall be the highest speed on any one channel. For example, a converter having double inputs and double outputs, not isolated from each other, shall be considered to have a ratio of input to output of 36 : 1 accurate to 0.5 degree, and a ratio of 1 : 1 accurate to 1 degree (see figure 6). Another converter having double isolated inputs and outputs shall be considered to have two separate 1 : 1 channels, each output being accurate to 0.5 degree (2 channel converter as specified on figure 1) (see 1.2.1).

3.8.6 Constant velocity accuracy. The constant velocity accuracy of the synchro converters shall be such that when tested as specified in 4.6.6, the errors shall not exceed those specified in the notes of the applicable figure (see figures 1 through 7 and 3.1.1).

3.8.7 Harmonic motion accuracy. The harmonic motion accuracy of the synchro converters shall be such that when tested as specified in 4.6.7, the errors shall not exceed those specified in the notes of the applicable figure (see figures 1 through 7 and 3.1.1).

3.8.8 Synchronization. Converters shall be self-synchronous from all displacements. When tested as specified in 4.6.8, the output synchros shall synchronize with the input synchros, stabilize, and attain the static accuracy specified in 3.8.5. Single- and double-speed converters shall attain synchronism within 1 degree of reference in not more than seven seconds. Triple-speed converters shall be self-synchronous; however, a time limit for achieving synchronism shall not apply.

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3.8.9 Phase shift. When tested as specified in 4.6.9, the converters shall operate as specified in 3.8.5 for all phase shifts between 15 degrees lag and 5 degrees lead caused by external synchro units in the synchro transmission system. For the purpose of this requirement, phase shift is defined as the electrical angular displacement of the fundamental component of the primary (stator) voltage from the secondary (rotor) voltage applied to the converter input.

3.8.10 Effect of magnetic fields. When tested as specified in 4.6.10, converters shall operate within the accuracy limits specified in 3.8.5, 3.8.6, and 3.8.7 while located in an ambient magnetic field of 5 gauss.

3.8.11 Power supply tolerance. When tested as specified in 4.6.11, converters shall conform to the static and dynamic accuracy requirements of 3.8.7.

3.8.12 Accelerated life. The static and dynamic accuracy of the converter shall be as specified in 3.8.5, 3.8.6, and 3.8.7 during each of the operational modes specified in 4.6.12.

3.8.13 Vibration. When tested as specified in 4.6.13, no damage shall occur and accuracy shall conform to the requirements of 3.8.5, 3.8.6, and 3.8.7.

3.8.14 Transients. When tested as specified in 4.6.14, no damage shall occur to the converters and dynamic accuracy shall conform to the requirements of 3.8.6 and 3.8.7.

3.8.15 Shock. When tested as specified in 4.6.15 no damage shall occur to the converter. During the test, dynamic accuracy shall conform to the requirements of 3.8.6 and 3.8.7. Following the test, static accuracy shall be as specified in 3.8.5.

3.8.16 Temperature and humidity. The tests specified in 4.6.16 shall be conducted to determine compliance with 3.8.5, 3.8.6, and 3.8.7.

3.8.17 Electromagnetic interference. When required (see 3.1.1), the converter shall be tested in accordance with MIL-STD-462, and shall conform to the requirements of MIL-STD-461.

3.8.18 Airborne noise. When required (see 3.1.1), converters shall be tested in accordance with 4.6.18, and shall conform to the requirements of MIL-STD-740 for grade C equipment.

3.8.19 Structureborne noise. When required (see 3.1.1), converters shall be tested as specified in 4.6.19, and shall conform to the requirements of MIL-STD-740 for type 3 equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection

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requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in table VI (see 6.4). Tests shall be conducted in the order specified in table VI.

4.3.1 Acceptance of various types. First article inspection of one type of converter will be considered for acceptance of other types of converters provided the following conditions are met:

(a) The enclosure is the same for all types of converters.
(b) The electronics module, drive motor, and switching network used for all type units is interchangeable or a single integral switch is used to convert from one type to the other.
(c) The unit passes all tests for one type operation and after conversion, (see (b) above), passes the tests indicated with a footnote for the other type operation (see table VI for first article inspection and table VII for quality conformance inspection).

4.4 Quality conformance inspection. Converters on the contract or order shall be subject to group A examination and test as specified in table VII. Group B inspection will be required by NAVSEA when the basic design of the equipment or when a vital part has been changed from that of the unit qualified. For example, a change from one design of amplifier to another design using different components shall be regarded as a basis design change. However, a change in gearing from 1- and 36-speed to 1- and 5-speed or from a 1- and 36-speed input and output to 1- and 36-speed input and 2- and 36-speed output shall not be regarded as a basic design change. One complete converter shall be subjected to each of the group B tests as specified in table VII. Any converter selected for group B tests shall not be used as a production converter.

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4.4.1 Rejection. The results of each test shall be compared with specification requirements. Failure to conform to this specification for any group A or B test shall be counted as a defect, and the converter shall not be offered for delivery.

4.5 General examination. The completed unit shall be given a thorough examination to determine that it conforms to the application specifications and drawings with respect to material, finish, workmanship, construction, assembly, dimensions, weight, and marking of identification and information plates. The examination shall be limited to those examinations that may be performed without disassembling the unit in such a manner that its performance, durability, or appearance would be affected. The examination shall include a check of operating controls, circuit functions, synchro connections as specified in 3.6.18, and test facilities. The examination shall determine that the equipment is as specified in 3.4 through 3.8.

4.6 Tests.

4.6.1 Dielectric withstanding voltage. Converters shall be tested in accordance with method 301 of MIL-STD-202 and the requirements of the applicable test potential shall be applied, between stator windings and case. The test potential shall be raised slowly (minimum time 3 seconds) from zero to 730 volts for 115-volt units, 200 volts for 26-volt units. The peak value of the test potential throughout this test shall not exceed 1.5 times the specified root mean square (rms) test potential. The test equipment employed shall differentiate between winding leaking current and surge discharge current. Following this test, the insulation resistance test shall be performed as specified in 4.6.2 and shall meet the requirements of 3.8.1.

4.6.2 Insulation resistance. Insulation resistance shall be measured by means of a device that applies 50 volts between each winding and case while indicating insulation resistance. Insulation resistance of the unit shall meet the requirements of 3.8.2.

4.6.3 Synchro connection. Converter shall be examined carefully, by disassembly as necessary, as specified in 3.8.3 and for comparison with the drawings for the equipment. Converter shall meet the requirements of 3.6.5.

4.6.4 Input operation. Converter shall be carefully examined as specified in 3.8.4 to ensure compliance with 3.6.3.

4.6.5 Overall static accuracy. Overall static accuracy (see 6.6.1) shall meet the requirements of 3.8.5. The accuracy shall be determined by turning the reference unit or synchro standard to six equally spaced positions in each direction. If the output synchro is a differential unit, the output's reference voltage shall come from a transmitter locked on zero. Figures 10 or 11 show possible test setups.

4.6.6 Constant velocity accuracy. Constant velocity accuracy (see 6.6.3) shall meet the requirements of 3.8.6. The constant velocity shall be checked for a minimum of one complete revolution of the converter's output synchro being tested. If the output synchro is a differential unit, the output's reference

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voltage shall come from a transmitter locked on zero. Figures 10 or 11 show possible test setups. The following are the speeds the reference unit shall be driven or simulated by the synchro standard during the test:

(a) Constant velocity of 32 degrees per second for converters having single or double outputs (see figures 1, 2, and 6).
(b) Constant velocity of 180 degrees per second into the 36X input or 30 degrees per second into the 1X input for triple output converters (see figures 3, 4, 5, and 7).

4.6.7 Simple harmonic motion accuracy. Simple harmonic motion (see 6.6.5) shall meet the requirements of 3.8.7. Figures 10 or 11 show possible test setups. Amplitude in simple harmonic motion is defined as the maximum angle displacement in each direction. The following are specifications for the input to the reference units:

(a) Simple harmonic motion of 40 degrees amplitude and a period of 8 seconds for converters having single output or double outputs (see figures 1, 2, and 6).
(b) Simple harmonic motion of 40 degrees amplitude and a period of 25 seconds for converters having triple outputs (see figures 3, 4, 5 and 7).

4.6.8 Synchronization. The converter's input shall be electrically coupled to a reference unit and energized. After the converter is aligned with the reference unit, the energizing voltage is turned off. If the converter's input consists of a CT, amplifier, and a servomotor, the amplifier may remain energized and the power to the servomotor turned off. The reference unit shall be rotated 177 degrees. The converter shall be re-energized. A timer shall be started simultaneously with the application of the voltage and stopped when synchronization occurs. Repeat the test with the reference unit set to 183 degrees. Unit shall meet the requirements of 3.8.8.

4.6.9 Phase shift. Synchro transmitters in the test director shall be loaded to produce the phase shifts specified in 3.8.9. The test director shall be positioned and turned as specified in 4.6.5, 4.6.6, and 4.6.7. The converter output error shall be determined as specified in 4.6.5.

4.6.10 Effect of magnetic fields. Static and dynamic accuracy tests (see 4.6.5, 4.6.6, and 4.6.7) shall be performed with the converter located in the specified magnetic field (see 3.8.10). Converter shall meet the requirements of 3.8.5.

4.6.11 Power supply tolerance. Converter shall be operated as specified in 4.6.7(a) or 4.6.7(b), as applicable, while energized from a type I or II power source in accordance with MIL-STD-1399, section 300, in which the voltage and frequency are at the upper tolerance limit (108 percent voltage and 105 percent frequency). The test shall be repeated with the voltage and frequency at the lower tolerance limit (92 percent voltage and 95 percent frequency). Converter shall meet the requirements of 3.8.11.

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4.6.12 Accelerated life test. Accelerated life test shall be conducted for a period of 250 hours and shall meet the requirements of 3.8.12. Minimum period of operation shall be 7 hours except as indicated. Conditions of operation shall be varied as specified in 4.6.7(a) or 4.6.7(b), as applicable. The 250-hour test shall be divided as shown in table VIII. Before starting the tests, an accuracy test shall be conducted and recorded. Upon completion of above test, an accuracy test shall be conducted and recorded. Immediately after the above endurance test is completed the insulation resistance test (see 4.6.2), shock test (see 4.6.15) and accuracy check test (4.6.5, 4.6.6, and 4.6.7) shall be conducted in the order specified.

4.6.13 Vibration. Converter shall be subjected to type I vibration in accordance with MIL-STD-167-1, except that the exploratory vibration test shall be omitted. During the vibration, the converter shall be operated as specified in 4.6.7(a) or 4.6.7(b), as applicable. Converter shall conform to the requirements of 3.8.13. Subsequent to the test, the converter shall be examined for damage and the static and dynamic accuracy shall be determined as specified in 4.6.5, 4.6.6, and 4.6.7.

4.6.14 Transients. With the converter operating as specified in 4.6.7(a) or 4.6.7(b), as applicable, a 1000 V pulse, in accordance with MIL-STD-1399, section 300, shall be superimposed on the supply voltage at the converter terminals. Following the pulse discharge, synchro dynamic errors shall be observed as specified in 4.6.6 and 4.6.7. The converter shall meet the requirements of 3.8.14.

4.6.15 Shock. Converter shall be tested in accordance with MIL-S-901 for grade A, class I, type B equipment. Converter shall be operated as specified in 4.6.7(a) or 4.6.7(b), as applicable. Following each blow, the converter shall be examined for damage. Following the test, the static accuracy shall be determined as specified in 4.6.5 to ensure compliance with 3.8.15.

4.6.16 Temperature and humidity. The following test conditions shall apply:

(a) Low temperature test shall be conducted with the unit in a chamber 4 hours at 40°F. Upon reaching the 4-hour point, the converter shall be energized and accuracy determined as specified in 4.6.5 to ensure compliance with 3.8.16.

(b) High temperature test shall be conducted with converter operating with a constantly varying input synchro signal of simple harmonic motion of 40 degrees amplitude and 8-second period (see 4.6.7). In addition, the heat dissipated from all sides of the converter shall be restricted by means of low heat-conducting bulkheads placed adjacent to the sides of the converter. Converter shall be operated uninterruptedly for 24 hours at not less than 149°F at a relative humidity of not less than 95 percent. Hot spot temperature shall not exceed 221°F which dictates that the temperature rise shall not exceed 75°F near the end of the test. The number and location of hot spot sensors shall be determined by scanning the various components in the unit to determine maximum hot spot temperatures. Not less than 3 hot spot sensors shall be attached to heat generating components in the converter. At anytime during and after the test, the converter shall meet the static anddynamic tests as specified in 4.6.5, 4.6.6, and 4.6.7.

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4.6.16.1 Internal examination. The converter and the internal mechanism shall be examined as specified in 3.13 and those requirements for 3.4, 3.5, and 3.6 that are amenable to visual inspection without disassembly of the mechanism. Any deterioration or damage that could in any manner prevent the converter from meeting functional operation or maintenance requirements during service life shall be cause for rejection.

4.6.17 Electromagnetic interference. When specified (see 3.1.1), converter shall be tested for electromagnetic interference in accordance with MIL-STD-462 and shall meet the requirements specified in 3.8.17.

4.6.18 Airborne noise. Converter shall be tested for airborne noise and shall meet the requirements specified in 3.8.18.

4.6.19 Structureborne noise. Converter shall be tested for structureborne noise and shall meet the requirements specified in 3.8.19.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation, packing, and marking. Equipment and accessories, repair parts, and publications shall be preserved level A or C; packed level A or B, as specified (see 6.2), and marked in accordance with MIL-E-17555.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. Converters are used for one or more of the following applications:

(a) To permit the operation of increased synchro loads.
(b) To permit the operation of a 400-Hz synchro load from a 60-Hz synchro bus or vice versa.
(c) To permit the transmission of a modified synchro signal.
(d) To permit the operation of a synchro load independent of the input synchro excitation.
(e) To minimize reflected errors on the input synchro signal bus.
(f) To provide a smooth and accurate transmission of synchro signals.

In addition, the attention of contractors is called to the fact that the equipment covered by this specification is intended for rigorous Naval shipboard service, where it is expected to withstand continuous use for long periods, under Military service conditions, without benefit of overhaul. The equipment is in each case a

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vital unit intended for extremely important usage by the Naval forces concerned. Failure at a critical moment results in a serious reduction of the battle efficiency of the ship involved. Emergency repairs afloat, or at distant bases, may result in the equipment being out of service during a critical shipboard period.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Type designation required (see 1.2 and 3.2.1).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) Fuse installation (see 3.6.4.1).
(e) Fuseholder installation (see 3.6.4.2).
(f) Energization of blown fuse indicator circuit (see 3.6.4.3).
(g) External electrical connections, specify connections as required (terminal board or connector) (see 3.6.5).
(h) If motoring need not be prevented (see 3.6.16.1).
(i) If synchro capacitors are to be provided as part of converter (see 3.6.17).
(j) When synchro electrical zero and equipment mechanical zero are not positioned to the same point (see 3.7.2).
(k) Levels of preservation and packing required (see 5.1).
(l) 1/18X accuracy to be 0.2 degree or 0.5 degree (see figures 3, 4, 5, and 7).

6.3 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.3.1 When technical manuals are required, MIL-STD-225 should be referenced for synchro maintenance information. If specified, loose leaf sheets should be provided as complete sets in the same format as for the manuals to completely cover maintenance and operation of the converter.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first ___ production items, a standard production item from the contractor's current inventory (see 3.1.2), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those

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bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 Definitions.

6.6.1 Static accuracy. Static accuracy is the largest static error at any position.

6.6.2 Static error. Static error is the difference between the converter's input angle and the converter's output angle times the input to output ratio while the converter is stationary.

6.6.3 Constant velocity accuracy. Constant velocity accuracy is the largest constant error at any position.

6.6.4 Constant velocity error. Constant velocity error is the difference between the converter's input angle and the converter's output angle times the input to output ratio. It is measured while the input is changing at a specified constant velocity.

6.6.5 Simple harmonic motion accuracy. Simple harmonic motion accuracy is the largest simple harmonic motion error at any position.

6.6.6 Simple harmonic motion error. Simple harmonic motion error is the difference between the converter's input angle and the converter's output angle times the input to output ratio. It is measured while the input is changing in a simple harmonic motion at a specified amplitude and period.

6.7 Classification. Converters will be identified by classification designation created as follows (see 3.1.1):

A	1	B	C
Number and gear ratio of input and output channels (see 1.2.1)	Accuracy of 1/18 speed output (see 1.2.2)	Input and output voltages and frequencies (see 1.2.3)	Size and type of output synchros (see 1.2.4)

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the converter shall be made by either a terminal board (type [illegible] or [illegible]

6.8 Subject term (key word) listing.

Degrees per revolution
Dynamic accuracy
Gear ratio
Multiple channel
Output device
Servocomponents
Servomotor
Servtorq
3 Speed output
Static accuracy

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Navy - SH

Review activities:
Navy - OS
DLA - GS

Preparing activity:
Navy - SH
(Project 6125-0238)

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TABLE I. Number and gear ratio of input and output channels.1/

Classification	Input type	Output type	Channels per panel	Gear ratios
A	Single	Single	1 channel	Direct coupled
B	Single	Single	2 independent	Direct coupled
C	Single	Single	3 independent	Direct coupled
D	Single	Single	1 channel	2 to 1
E	Single	Single	2 independent	2 to 1
F	Single	Single	1 channel	1 to 2
G	Single	Single	2 independent	1 to 2
H	Single	Triple	1 channel	1 : 1/18, 1 & 36
I	Single	Triple	1 channel	36 : 1/18, 1 & 36
J	Double	Triple	1 channel	1,36 : 1/18, 1, 36
K	Double	Double	1 channel	1 & 36 : 1 & 36
L	Double	Triple	1 channel	1,36 : 1/18, 1, 36

1/ The number of output channels also determines the number of output devices.

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TABLE II. Input and output voltages and frequencies.

TYPE	Input	Output
A	115 volts - 60 hertz	115 volts - 60 hertz
B	115 volts - 60 hertz	115 volts - 400 hertz
E	115 volts - 400 hertz	115 volts - 60 hertz
F	26 volts - 400 hertz	115 volts - 400 hertz
H	26 volts - 400 hertz	115 volts - 60 hertz
J	115 volts - 400 hertz	115 volts - 400 hertz
M	115 volts - 60 hertz	26 volts - 400 hertz
N	115 volts - 400 hertz	26 volts - 400 hertz
P	26 volts - 400 hertz	26 volts - 400 hertz

TABLE III. Size and type of output synchros.

Group	Part no.	Output device type
A	23CX	Synchro control transmitter.
B	31CX	Synchro control transmitter.
C	37CX	Synchro control transmitter.
D	23CDX	Synchro control differential transmitter.
E	37CDX	Synchro control differential transmitter.
F	23TRX	Synchro torque receiver-transmitter.
G	31TRX	Synchro torque receiver-transmitter.
H	37TRX	Synchro torque receiver-transmitter.
I	23TDX	Synchro torque differential transmitter.
J	31TDX	Synchro torque differential transmitter.
K	37TDX	Synchro torque differential transmitter.

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TABLE IV. Maximum permissible sizes of output synchros for various converter configurations.

Converter configuration	MIL-C-24105/xx number	Maximum load-output synchro(s)	
		Size	Types
Single channel servo direct-coupled to output synchro	MIL-C-24105/1 through MIL-C-24105/27	23, 31 or 37	CX, CDX, TRX, or TDX
Single channel 360 deg. per rev. input, 180 deg. per rev. output	MIL-C-24105/28 through MIL-C-24105/45	23	CX, CDX, TRX, or TDX
Single channel 180 deg. per rev. input, 360 deg. per rev. output	MIL-C-24105/46 through MIL-C-24105/63	23	CX, CDX, TRX, or TDX
One speed input (1 servo & 1 CT), three speed output	MIL-C-24105/64 through MIL-C-24105/72	23	CX or TRX
One speed input, three speed output	MIL-C-24105/73 through MIL-C-24105/81	23	CX or TRX
Two speed input (1 servo & 2 CT's), three speed output	MIL-C-24105/82 through MIL-C-24105/91	23	CX or TRX
Two speed input, two speed output	MIL-C-24105/92 through MIL-C-24105/99	23	CX or TRX
Two speed input (two servos), three speed output	MIL-C-24105/100 through MIL-C-24105/108	23	CX or TRX

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TABLE V. Representative values of synchro capacitors or specific synchro units.

Synchro type	Capacitance per leg in microfarads (uF) 1/
15CT6	0.34
18CT6	.24
23CT6	.34
11CT4	.036
15CT4	.022
18CT4	.014
23CT4	.014

1/ Capacitance per leg is defined as meaning the capacitor element in each of the delta may have the value indicated above for a specific synchro unit, upon completion of the calculations specified in NAVSEA DDS 430-2.

Example: Type 15CT6, 0.34 uF per leg

$$C_{12} = C_{13} = C_{23} = 0.34\mu F$$

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TABLE VI. First article inspection.

Inspection	Requirement	Test
General examination	3.4, 3.5, 3.6, 3.7, 3.8	4.5
Dielectric withstanding voltage	3.8.1	4.6.1
Insulation resistance	3.8.2	4.6.2
Synchro connections	3.8.3	4.6.3
Input operation	3.8.4	4.6.4
Accuracy and oscillations 1/	3.8.5, 3.8.6, 3.8.7	4.6.5, 4.6.6, 4.6.7
Synchronization 1/	3.8.8	4.6.8
Phase shift	3.8.9	4.6.9
Effect of magnetic fields	3.8.10	4.6.10
Power supply tolerance 1/	3.8.11	4.6.11
Accelerated life	3.8.12	4.6.12
Vibration	3.8.13	4.6.13
Transients	3.8.14	4.6.14
Shock	3.8.15	4.6.15
Temperature and humidity	3.8.16	4.6.16
Electromagnetic interference 1/	3.8.17	4.6.17
Airborne noise	3.8.18	4.6.18
Structureborne noise	3.8.19	4.6.19

1/ See 4.3.1 (c).

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TABLE VII. Quality conformance inspection.

Inspection	Requirement	Test
Group A		
General examination	3.4, 3.5, 3.6, 3.7, 3.8	4.5
Synchro connections	3.8.3	4.6.3
Input operation	3.8.4	4.6.4
Accuracy and oscillations	3.8.5, 3.8.6, 3.8.7	4.6.5, 4.6.6, 4.6.7
Synchronization	3.8.8	4.6.8
Phase shift	3.8.9	4.6.9
Group B		
General examination	3.4, 3.5, 3.6, 3.7, 3.8	4.5
Power supply tolerance 1/	3.8.11	4.6.11
Electromagnetic interference	3.8.17	4.6.17
Airborne noise	3.8.18	4.6.18
Structureborne noise	3.8.19	4.6.19

1/ See 4.3.1 (c).

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TABLE VIII. Accelerated life test.

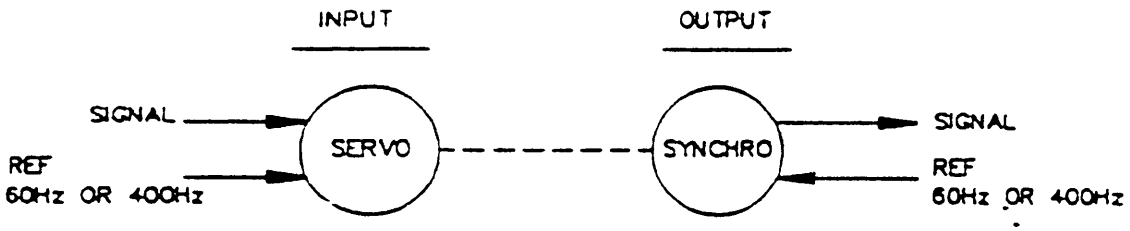
Hours 1/	Ambient temperature during test	Test conditions	Angle of tilt
0 - 25	75°F nominal (room temperature)	Constantly varying	Normal
After 25	Vibration test (4.6.13), accuracy check test, and short test runs (1/2 hour each) at undervoltage & underfrequency and at overvoltage and overfrequency (see 4.6.11)		
25 - 85 continuous	40°F	Constantly varying	Normal
85 - 150 continuous	149°F at 95 ± 4 percent relative humidity and then chamber shut down for alternate 1-hour periods. Hot spots, internal air and ambient temperature shall be continuously scanned.	Constantly varying	Normal
150 - 175	75°F nominal (room temperature)	Constantly varying	Inclined 60 deg's forward from vertical
175 - 200	75°F nominal (room temperature)	Constantly varying	Inclined 60 deg's back from vertical
200 - 225	75°F nominal (room temperature)	Constantly varying	Inclined 60 deg's right from vertical
225 - 250	75°F nominal (room temperature)	Constantly varying	Inclined 60 deg's left from vertical

1/ Time periods may be varied to suit laboratory test schedule.

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Office of Technology and Standards

CONVERTERS, SYNCHRO SIGNAL (PANEL MOUNT)



NOTES:

1. OUTPUT SYNCHRO MAY BE TYPE CX, CDX, TDX, OR TRX OF SIZES 23, 31, OR 37.

2.

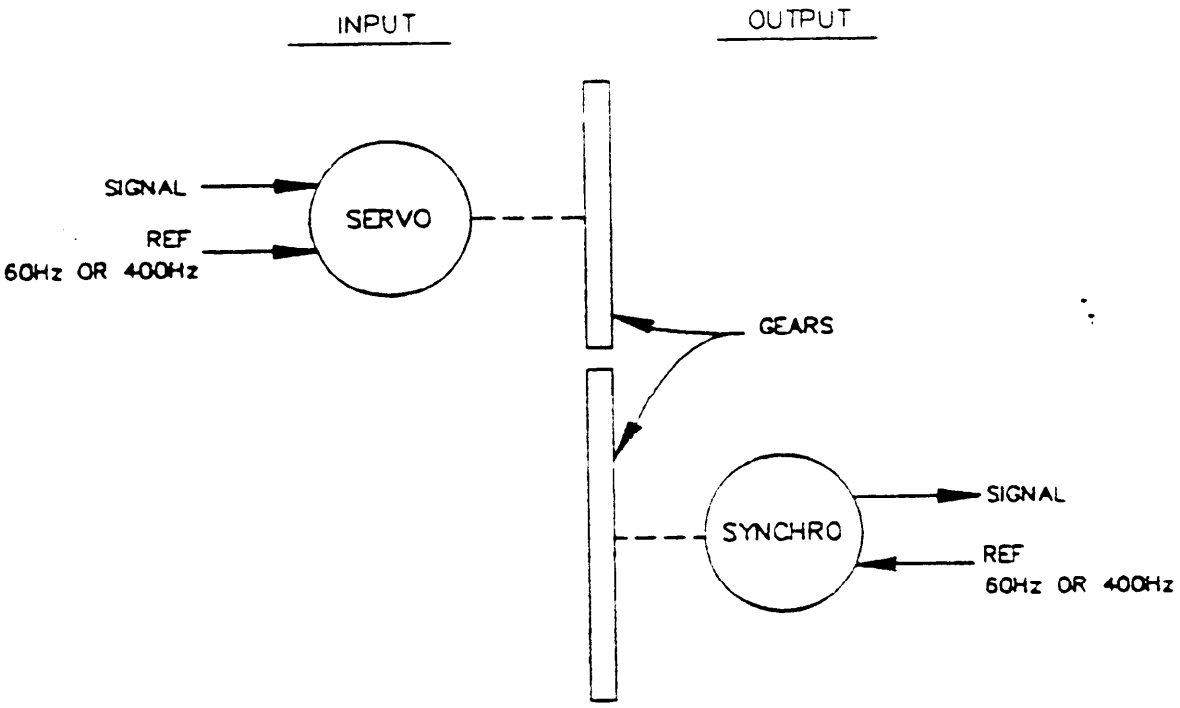
IF INPUT IS	OUTPUT IS	ACCURACY IS
1X	1X	0.5 DEGREE
2X	2X	0.5 DEGREE
36X	36X	0.5 DEGREE

3. UP TO 3 INDEPENDENT CHANNELS MAY BE PROVIDED IN A SINGLE CONVERTER PANEL, PROVIDED THE OUTPUT SYNCHROS ARE SIZE 23, AND NOT MORE THAN TWO OF THEM ARE DIFFERENTIALS.

FIGURE 1. Single input-single output, one, two or three independent or isolated channels, direct coupled types.

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CONVERTERS, SYNCHRO SIGNAL (PANEL MOUNT)



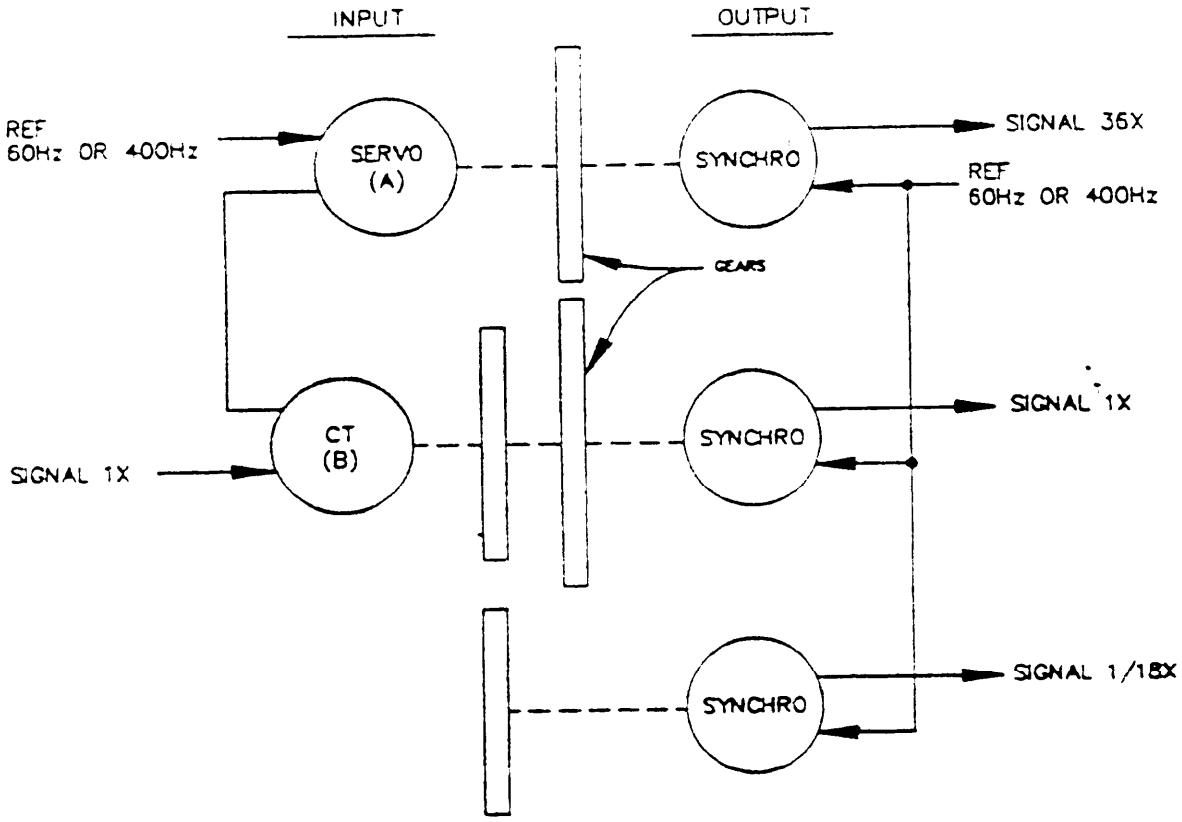
NOTES:

1. OUTPUT SYNCHRO MAY BE TYPE CX, CDX, TDX, OR TRX OF SIZE 23 ONLY.
2. TWO CHANNELS MAY BE PROVIDED IN A ONE PANEL CONVERTER.
3. ACCURACY IS 1.0 DEGREE.

FIGURE 2. Single input-single output, one or two independent or isolated channels, 2 to 1 or 1 to 2 gear ratio types.

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CONVERTERS, SYNCHRO SIGNAL (PANEL MOUNT)



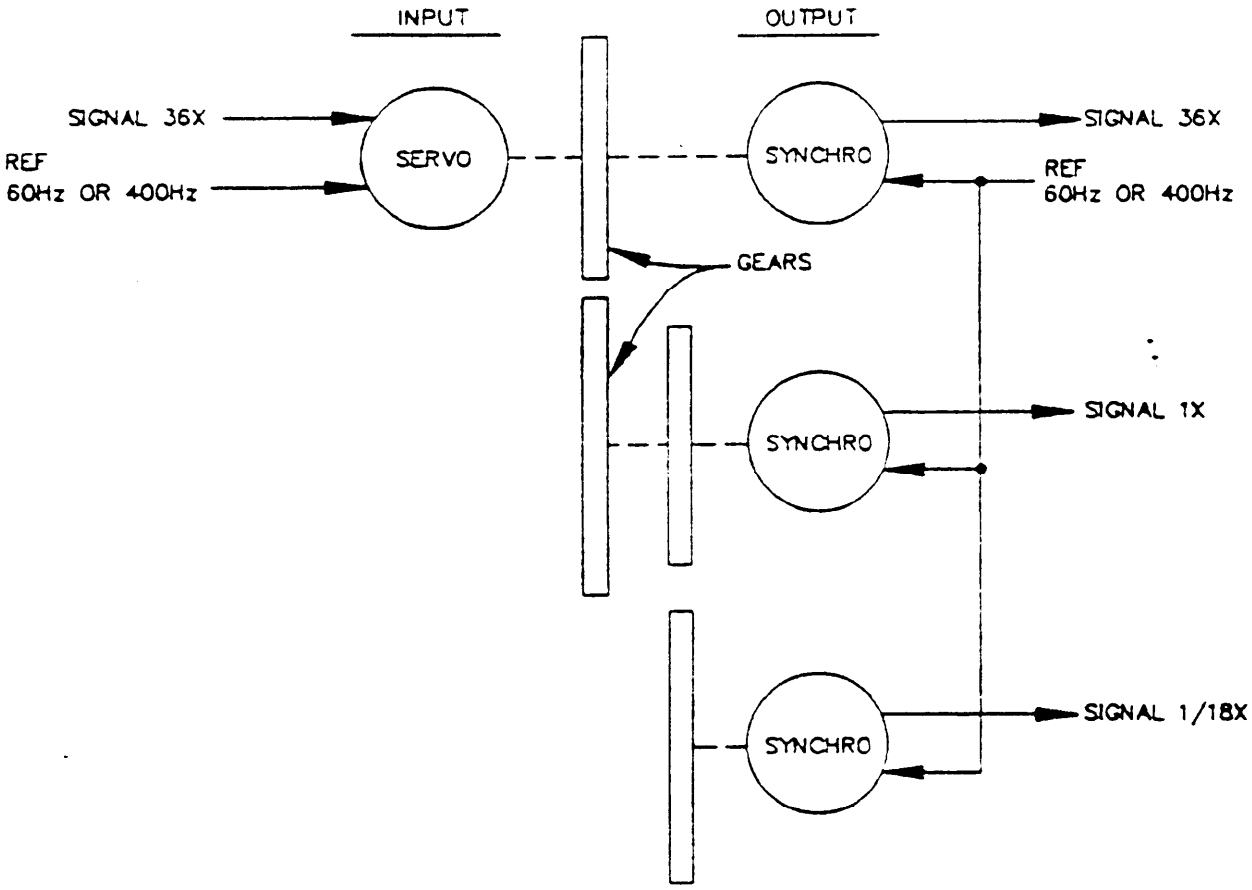
NOTES:

1. SERVO(A) AND CT(B) MAY BE A SPLIT VERSION OF THE STANDARD 23ST64A.
   (A) CONTAINS D.C. TORQUE MOTOR AND ELECTRONICS
   (B) IS A CT ONLY
2. OUTPUT SYNCHROS MAY BE CX, CDX, OR TRX, SIZE 23.
3. DIRECT-DRIVE NO-SPEED SWITCHING NETWORK IS REQUIRED.
4. CONVERTER SIZE IS FOR ONE PANEL.
5. 36X ACCURACY = 36 DEGREES.
   1X ACCURACY = 1 DEGREE.
   1/18X ACCURACY = 0.2 DEGREE WITH ANTI-BACKLASH GEARS, OR
   0.5 DEGREE WITHOUT ANTI-BACKLASH GEARS.

FIGURE 3. Single input-triple output, non-isolated channel, 1 to 1/18, 1 and 36 gear ratio type.

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CONVERTERS, SYNCHRO SIGNAL (PANEL MOUNT)



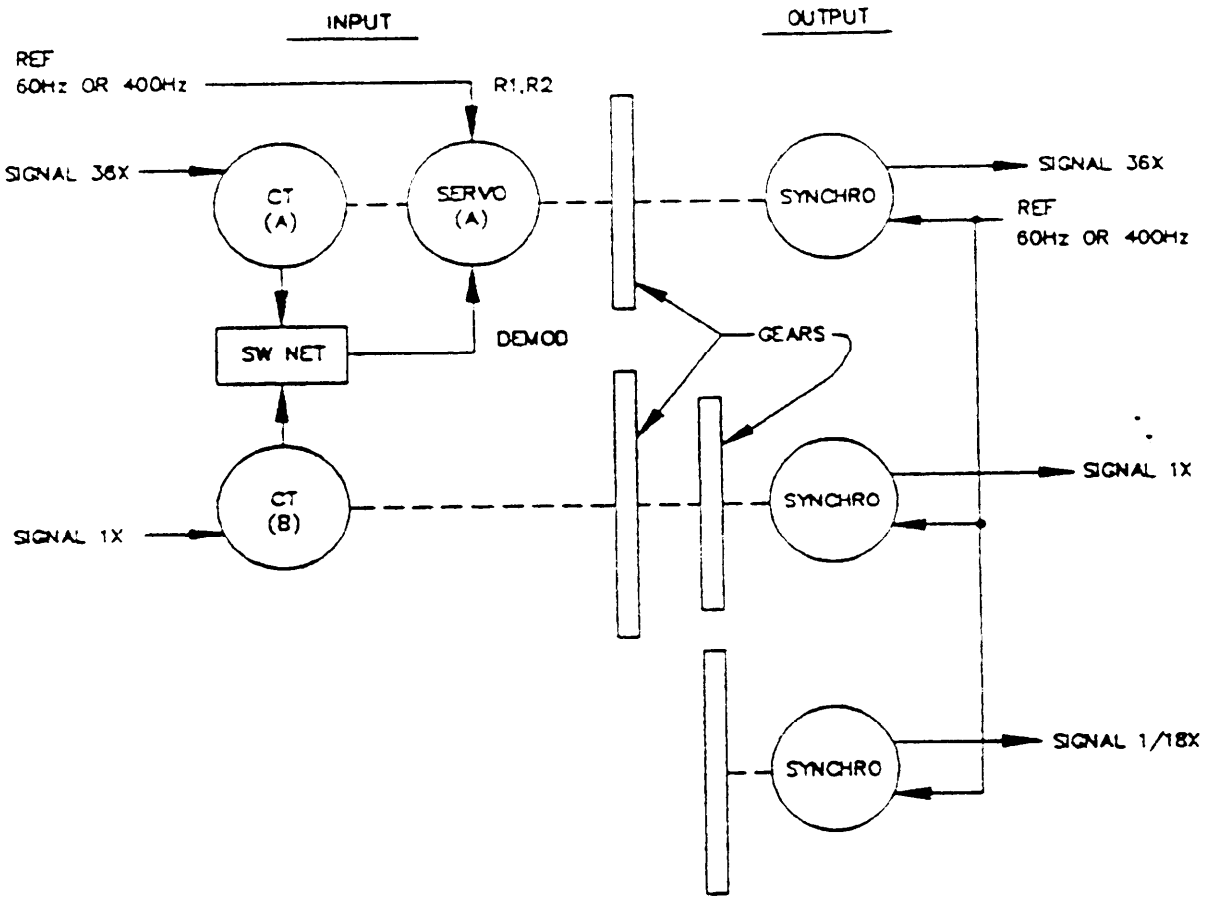
NOTES:

1. OUTPUT SYNCHROS MAY BE CX OR TRX, SIZE 23.
2. DIRECT-DRIVE NO-SPEED SWITCHING NETWORK IS REQUIRED.
3. CONVERTER SIZE IS FOR ONE PANEL.
4. 36X ACCURACY = 1 DEGREE.
   1X ACCURACY = 0.5 DEGREE.
   1/18X ACCURACY = 0.2 WITH ANTI-BACKLASH GEARS, OR
   0.5 DEGREE WITHOUT ANTI-BACKLASH GEARS.

FIGURE 4. Single input-triple output, non-isolated channels 36 to 1/18, 1 and 36 gear ratio type.

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CONVERTERS, SYNCHRO SIGNAL (PANEL MOUNT)



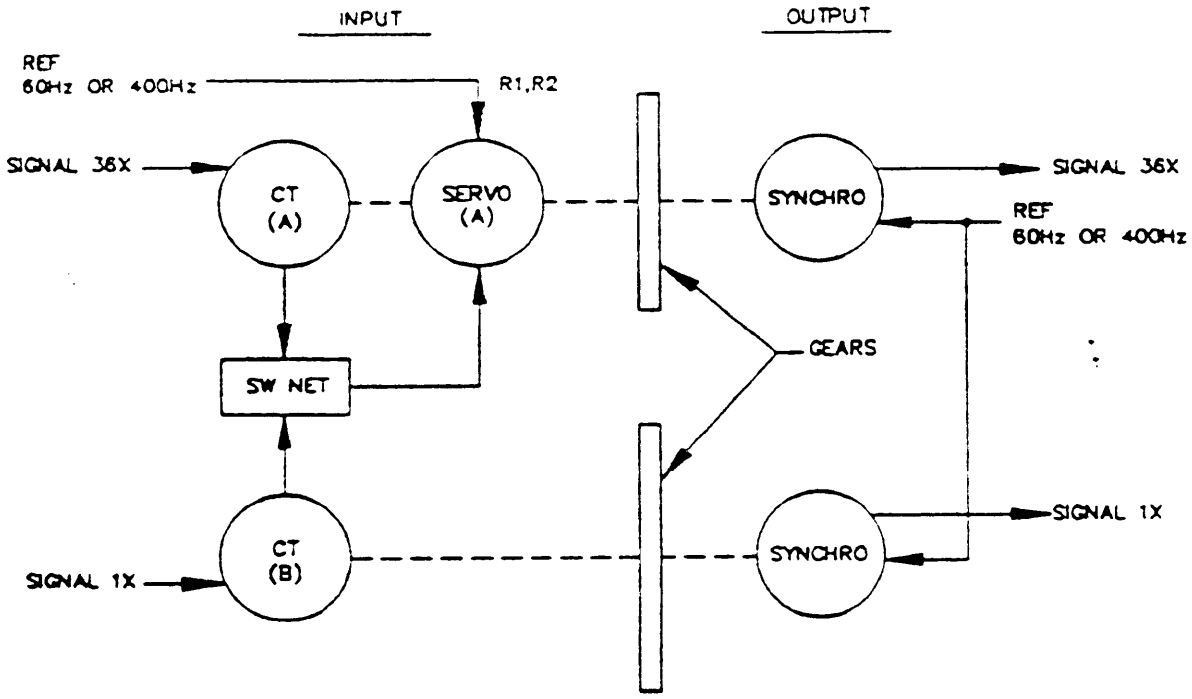
NOTES:

1. CT(A) AND SERVO(A) MAY BE A SERVTORQ (23ST64A) WITH THE DEMODULATOR INPUT LEADS BROUGHT OUT.
2. CT(B) IS A STANDARD CT HOUSED SEPARATELY.
3. CT(A) AND (B) ROTOR LEADS CONNECT TO INPUT OF SPEED SWITCHING NETWORK. OUTPUT OF SPEED SWITCHING NETWORK CONNECTS TO ST DEMODULATOR INPUT (DEM).
4. OUTPUT SYNCHROS MAY BE CX OR TRX, SIZE 23.
5. CONVERTER SIZE IS FOR ONE PANEL.
6. 36X ACCURACY = 1 DEGREE.
   1X ACCURACY = 0.5 DEGREE.
   1/18X ACCURACY = 0.2 DEGREE WITH ANTI-BACKLASH GEARS, OR
   0.5 DEGREE WITHOUT ANTI-BACKLASH GEARS.

FIGURE 5. Double input-triple output, non-isolated channel, 1 and 36 to 1/18, 1 and 36 gear ratio type.

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CONVERTERS, SYNCHRO SIGNAL (PANEL MOUNT)



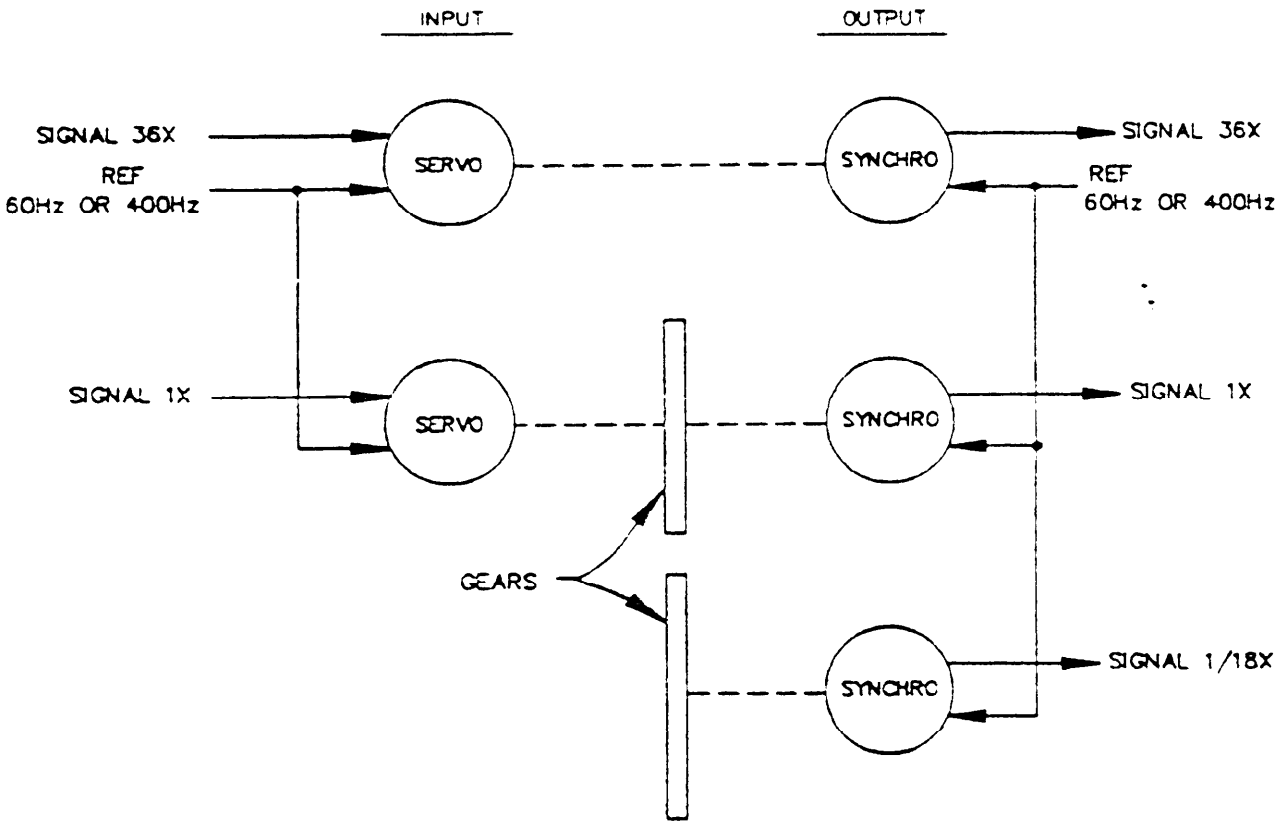
NOTES:

1. SERVO(A) AND CT(A) MAY BE A SERVTORQ (23ST64A) WITH THE DEMODULATOR INPUT LEADS BROUGHT OUT.
2. CT(B) IS A STANDARD CT HOUSED SEPARATELY.
3. CT(A) AND (B) ROTOR LEADS CONNECT TO INPUT OF SPEED SWITCHING NETWORK. OUTPUT OF SPEED SWITCHING NETWORK CONNECTS TO ST DEMODULATOR INPUT (DEM).
4. OUTPUT SYNCHROS MAY BE CX OR TRX, SIZE 23.
5. CONVERTER SIZE IS FOR ONE PANEL.
6. 36X ACCURACY = 1 DEGREE.
   1X ACCURACY = 0.5 DEGREE.

FIGURE 6. Double input-double output, non-isolated channels, 1 and 36 to 1 and 36 gear ratio type.

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CONVERTERS, SYNCHRO SIGNAL (PANEL MOUNT)

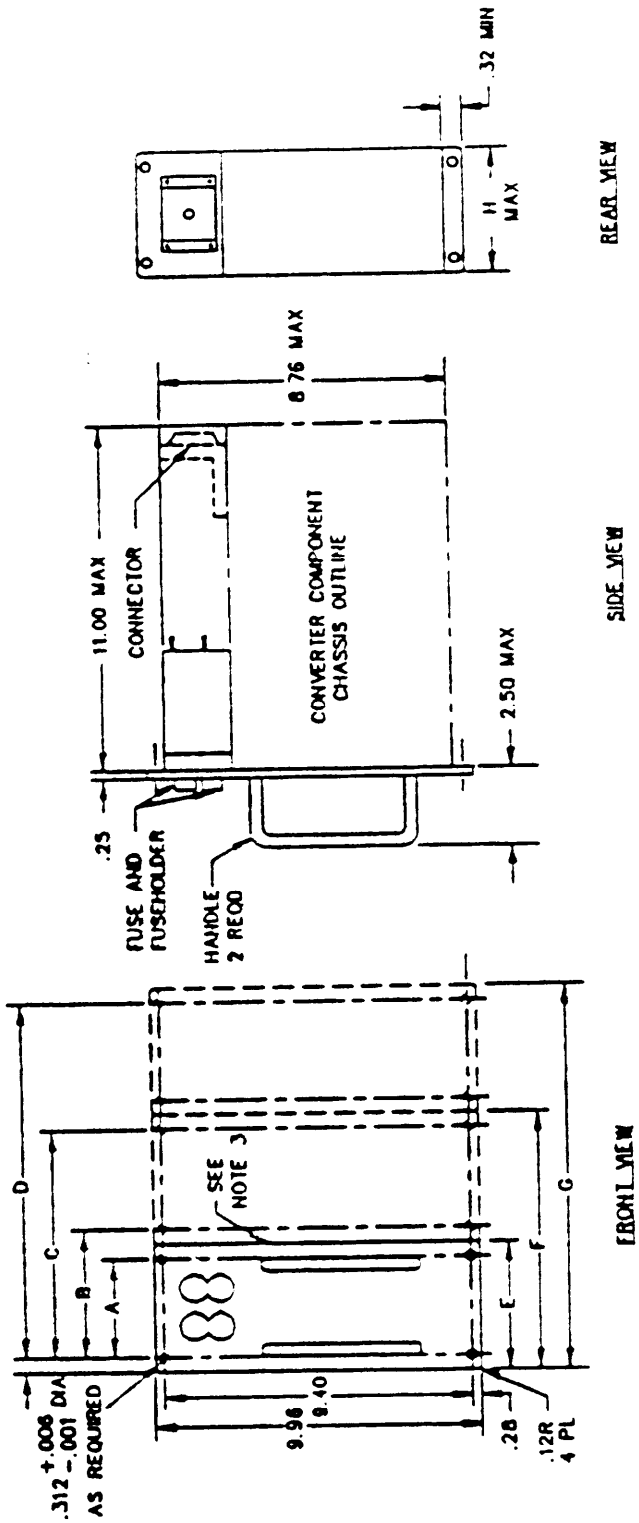


NOTES:

1. OUTPUT SYNCHROS MAY BE CX OR TRX, SIZE 23.
2. NO-SPEED SWITCHING NETWORK IS PROVIDED.
3. CONVERTER SIZE IS FOR ONE PANEL.
4. 36X ACCURACY = 0.5 DEGREE.
   1X ACCURACY = 1 DEGREE.
   1/18X ACCURACY = 0.2 DEGREE WITH ANTI-BACKLASH GEARS, OR
   0.5 WITHOUT ANTI-BACKLASH GEARS.

FIGURE 7. Double input-triple output, independent or isolated channels, 1 and 36 to 1/18, 1, and 36 gear ratio type.

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Converter front panel width	Dimensions Mounting holes				Panel width				
	A	B	C	D	E	F	G	H
1 Panel	3.060	-----	-----	-----	3.96	-----	-----	3.88
2 Panel	3.060	4.000	7.060	-----	-----	7.96	-----	7.88
3 Panel	3.060	4.000	7.060	11.060	-----	-----	11.96	11.88

NOTES:

1. All dimensions are in inches.
2. Tolerances are 0.XX ± 0.02, 0.XXX ± 0.010.
3. Front panel to be painted in accordance with system 37 of MIL-STD-1303.

FIGURE 8. Converter envelope dimensioning requirements.

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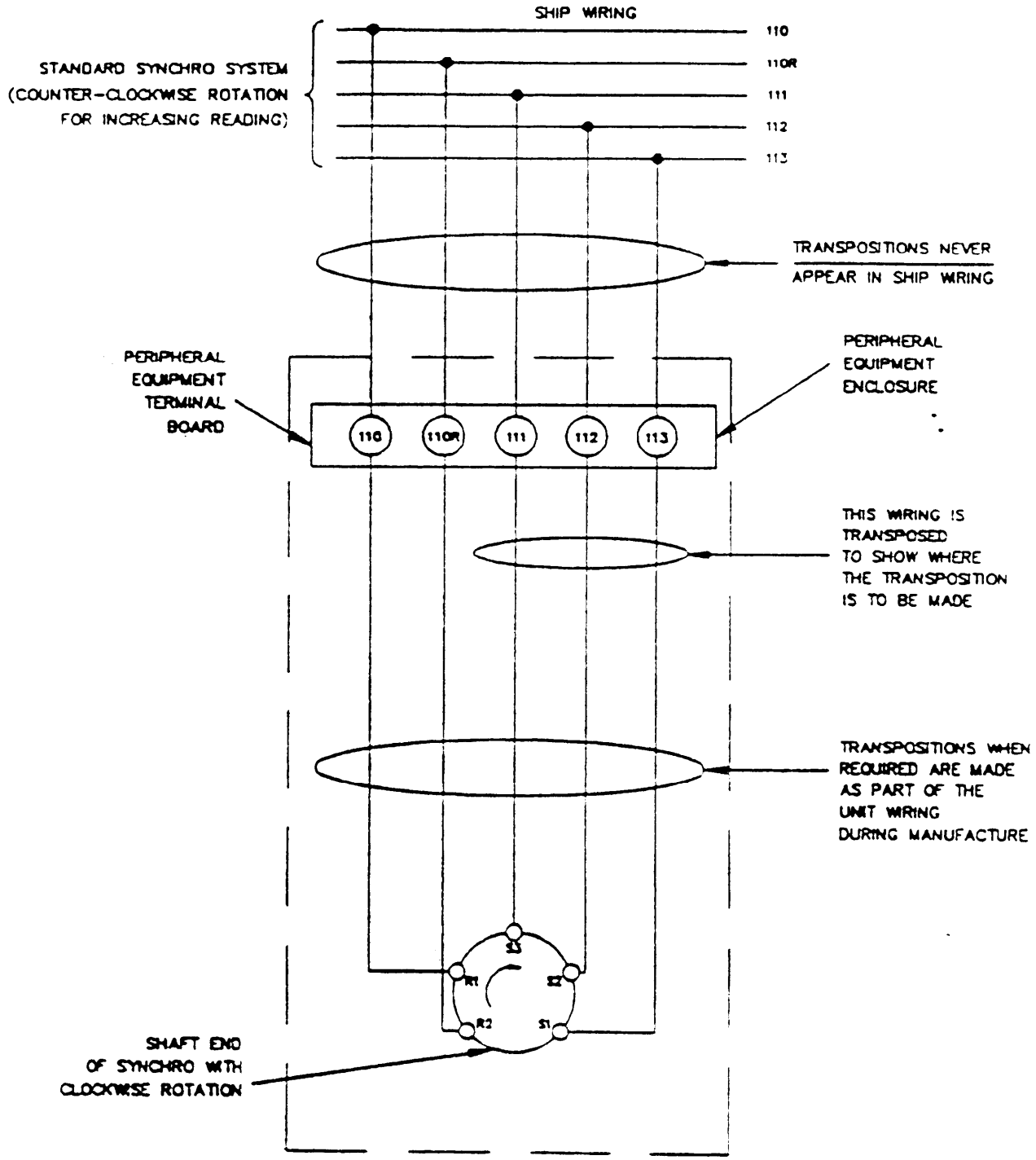


FIGURE 9. Connections for clockwise rotating synchro.

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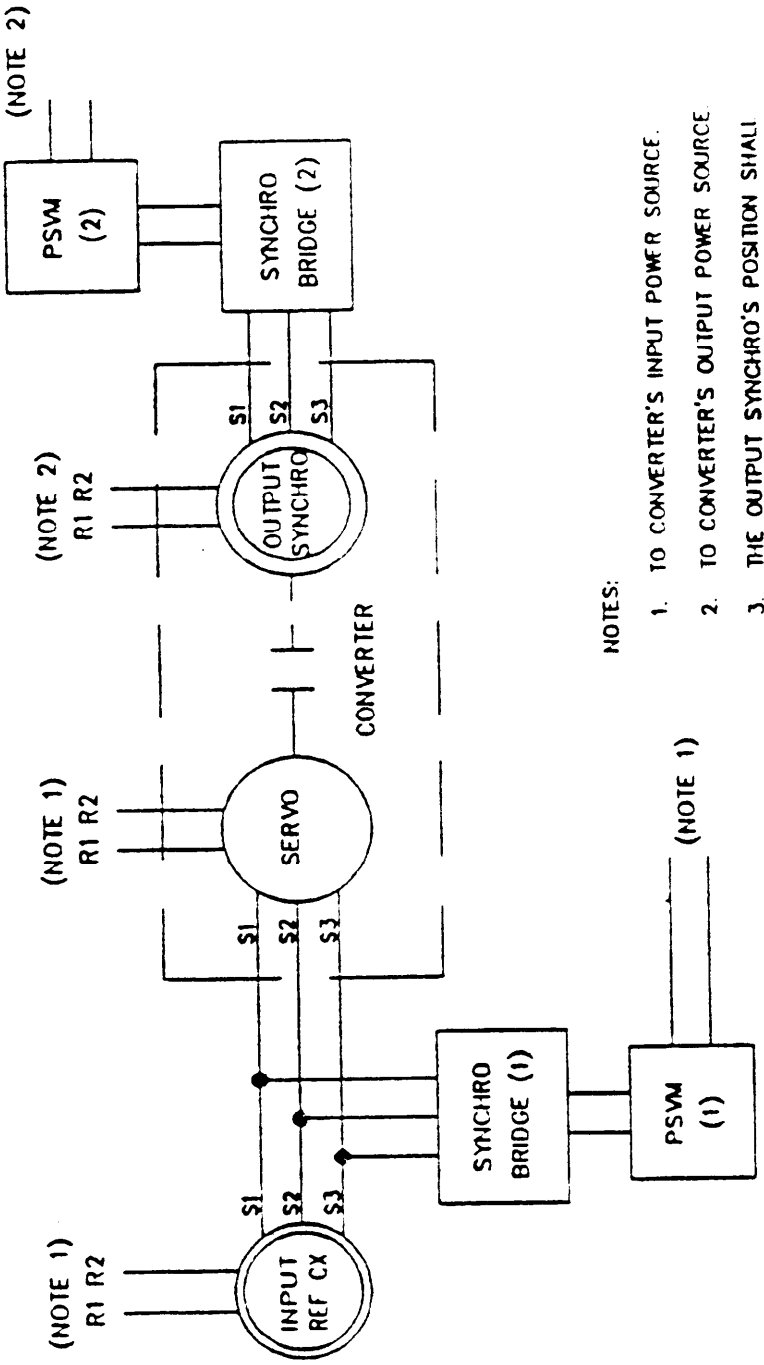
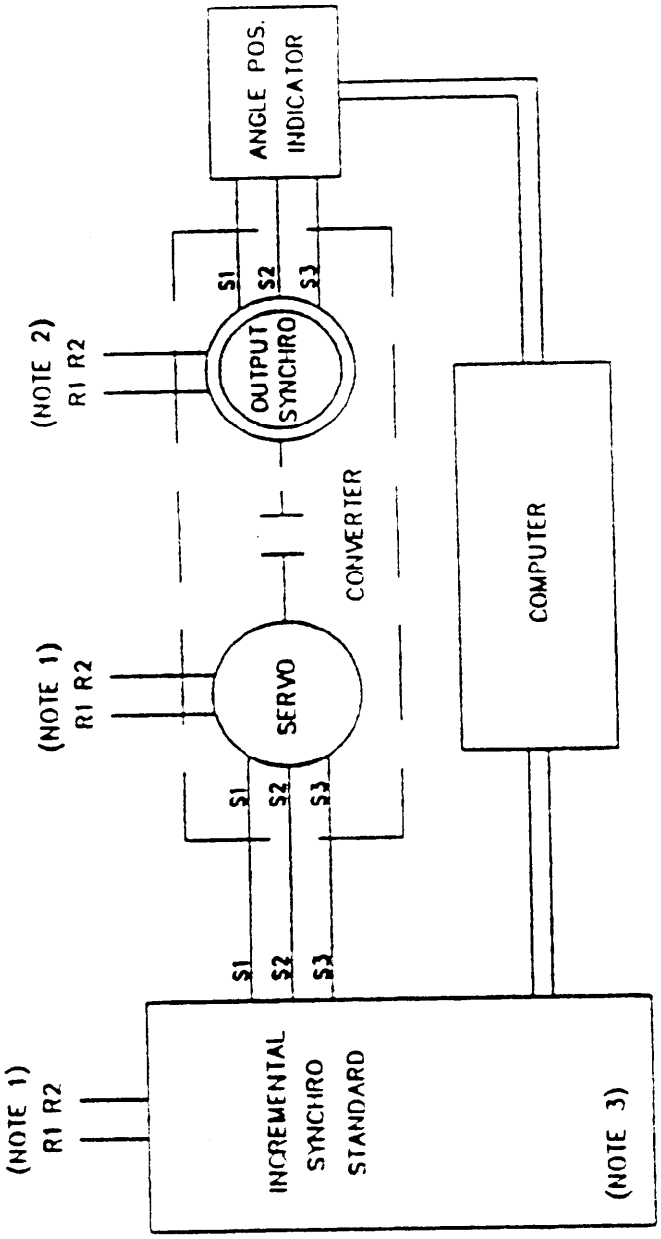


FIGURE 10. Accuracy test circuit for converters using synchro bridges and phase sensitive voltmeters.

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NOTES:

1. TO CONVERTER'S INPUT POWER SOURCE.
2. TO CONVERTER'S OUTPUT POWER SOURCE.
3. INCREMENTAL SYNCHRO STANDARD SHALL HAVE A MAXIMUM RESOLUTION OF 0.01 DEGREE AND A DIGITAL INPUT CONTROLLED BY THE COMPUTER.
4. THE OUTPUT SYNCHRO'S POSITION SHALL BE MULTIPLIED BY THE GEAR RATIO AND SUBTRACTED FROM THE INPUT SYNCHRO'S POSITION TO OBTAIN THE ERROR.

FIGURE 11. Accuracy test circuit for converters using an incremental synchro standard, angle position indicator, and a computer.

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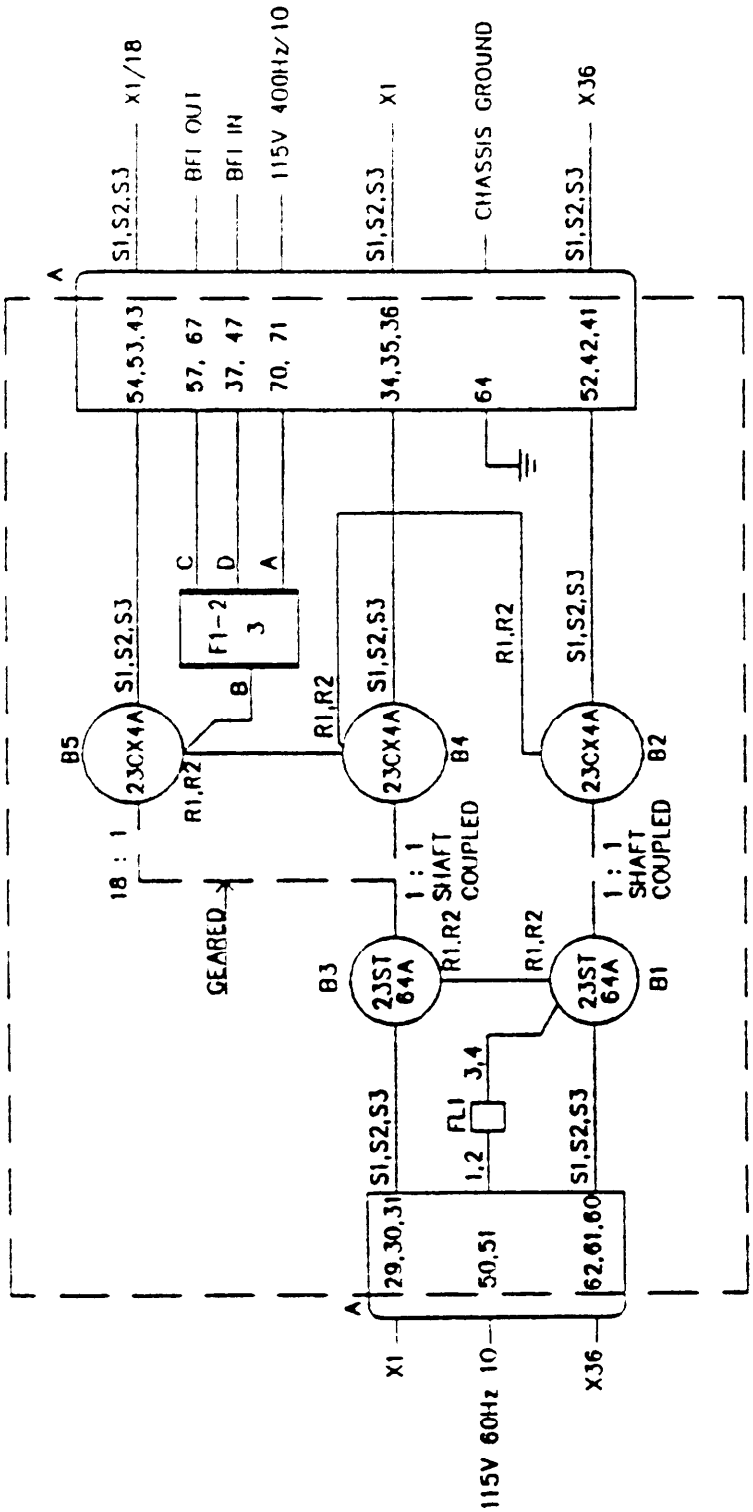


FIGURE 12. Example synchro signal converter wiring diagram, using MIL-C-28748 connector.

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